Project Proposal

Human Action Recognition for Human-Robot Interaction (HRI)

Proposer: **Ultra Vision**
Proposer Member: Alessandro

May 26, 2026

1 Introduction and Project Goal

Human-Robot Interaction (HRI) relies heavily on computer vision systems to understand the surrounding environment and the behavior of human operators. In collaborative robotics and smart environments, a system must rapidly recognize human actions over a continuous execution stream to ensure operational safety (e.g., triggering automated safety protocols or collision avoidance) and provide contextual assistance.

The goal of this project is to develop a robust computer vision system capable of analyzing image sequences to detect and classify six fundamental human actions: *walking, jogging, running, boxing, hand waving*, and *hand clapping.*

The required system must process the temporal sequences and provide high-level kinematic and semantic information at each frame. Specifically, the system should be able to:

- Isolate the human actor from the background using a robust segmentation framework.
- Track the subject's dynamic bounding box across continuous temporal sequences.
- Analyze spatial-temporal movement dynamics to extract motion features.
- **Classify the entire 40-frame sequence with a single global action label** and display real-time labeling alongside the bounding box visualization.

2 Case Studies and Simplifications

As case studies for the development and validation of the required system, various video sequences from the standard KTH Action Recognition Dataset are considered. To evaluate the true generalization capabilities of the architecture, the system must be robust across 4 distinct environmental scenarios: standard outdoors ($d1$), outdoors with scale and zoom variations ($d2$), outdoors with clothing variations ($d3$), and indoor environments with shifting lighting conditions ($d4$). The framework must successfully ignore any static background clutter and focus entirely on the human actor.

In developing the project, the following technical simplifications can be assumed:

- The camera pose remains strictly stationary during each video clip (static background, absence of ego-motion).
- Only a single human actor is present in the playing field/scene at any given time.
- While the core feature extraction and classification pipeline strictly relies on classical algorithmic Computer Vision, a **Hybrid Pre-processing Layer** (e.g., integrating lightweight pre-trained pose estimation models like MediaPipe) is permitted as an architectural fallback strategy solely to ensure silhouette extraction stability under severe scale ($d2$) or clothing ($d3$) variations.

3 Dataset Stratification, Ground Truth, and Pipeline

The foundational video data is sourced from the official **KTH Action Recognition Dataset** (publicly available at `https://www.csc.kth.se/cvap/actions/`). However, rather than using raw, uncurated video files directly from this source, the project defines a highly structured data processing and evaluation strategy based on pre-extracted image frame sequences to ensure absolute compatibility within the course's Virtual Machine environment.

1. **Stratified Selection (Image Sequences):** A meticulously balanced dataset of **72 action sequences** is defined (12 sequences per action class). To eliminate environmental and lighting biases, the sequences are symmetrically sampled from the four KTH scenarios (3 sequences from each of the $d1, d2, d3, d4$ environments per class).

2. **Temporal Window Logic:** For each of the 72 sequences, instead of a raw video file, the dataset provides exactly **40 pre-extracted continuous frames** saved as image files. This 40-frame window (representing approx. 1.6 seconds of execution at 25 FPS) contains the exact duration required to capture at least one complete kinematic cycle (e.g., a full gait stride or a complete hand-clap). This removes any video codec dependency and allows immediate frame-by-frame processing.

3. **Ground Truth Annotations:** To evaluate the system, two ground truth components are provided for each 72-sequence folder:

   - **Sequence Label:** A single global classification label (integer from 1 to 6) representing the action class for the entire 40-frame sequence.
   - **Median Frame Bounding Box:** To evaluate localization without requiring massive annotation overhead, the **median frame (the 20th frame)** of each sequence is selected as the reference.

   A corresponding `.txt` file merges these two pieces of information, containing both the sequence label (<class_id>) and the exact ground truth bounding box coordinates for this frame using the standard format:

   ```
   <class_id> <x_center> <y_center> <width> <height>
   ```

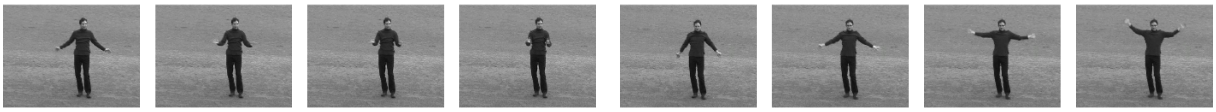
4. **Custom Processed Dataset Availability:** The complete validation dataset, structured into sequence folders containing the 40 frames and their respective ground truth `.txt` files, is hosted in the following shared Google Drive folder:
   `https://drive.google.com/drive/folders/13tP1OnK2clin47TbCLNlUzBvLhoLGaER?usp=sharing`

4 Technical Challenges and Ambiguity Resolution

The processing framework is specifically designed to methodically tackle the two historical ambiguities inherent to motion benchmark datasets:

4.1 Challenge I: Motion Directionality (Handclapping vs. Handwaving)

These two actions share near-identical static silhouettes, similar bounding box aspect ratios, and virtually zero global centroid displacement. To resolve this ambiguity, the feature extraction framework must isolate the directionality of local motion vectors within the 40-frame window, separating horizontal convergence/divergence (clapping) from vertical and angular oscillations above the shoulder line (waving).



(a) Handclapping Local Motion Flow (b) Handwaving Local Motion Flow

Figure 1: Methodological approach for Challenge I: isolating local vector directionality.

4.2 Challenge II: Kinematic Intensity (Jogging vs. Running)

Jogging and running involve identical leg/arm mechanics and horizontal translation across the scene. To successfully separate them, the system must quantify the kinematic intensity through two specific metrics: measuring the magnitude of the global translation vector (Centroid Velocity) and analyzing the temporal frequency of the silhouette's periodic deformation (Stride Frequency).

(a) Jogging Displacement Pattern (b) Running Displacement Pattern

Figure 2: Methodological approach for Challenge II: evaluating global velocity and stride period.

5 Performance Metrics

To assess the performance and robustness of the developed system against the annotated ground truth, standard computer vision and machine learning metrics will be implemented:

5.1 Localization Metrics (Bounding Box)

- **Intersection over Union (IoU):** Computed specifically on the designated median frame (20th frame) of each sequence to measure the spatial overlap accuracy between the dynamically predicted bounding box and the provided ground truth annotations.
- **Mean IoU (mIoU):** The mathematical average of IoU scores across all 72 test sequences to assess overall bounding box localization stability.

5.2 Classification Metrics (Category)

- **Global Accuracy:** The exact percentage of correctly classified sequences (assigning a single correct global action label per 40-frame folder) over the total 72 sequences.
- **F1-Score:** Evaluated for each of the six classes to provide a balanced measurement of Precision and Recall.
- **Confusion Matrix:** Mandatory analytical tool used to measure and visualize cross-class misclassifications, specifically monitoring the expected overlap between jogging and running.

6 Input Image Sequences (Starting Dataset)

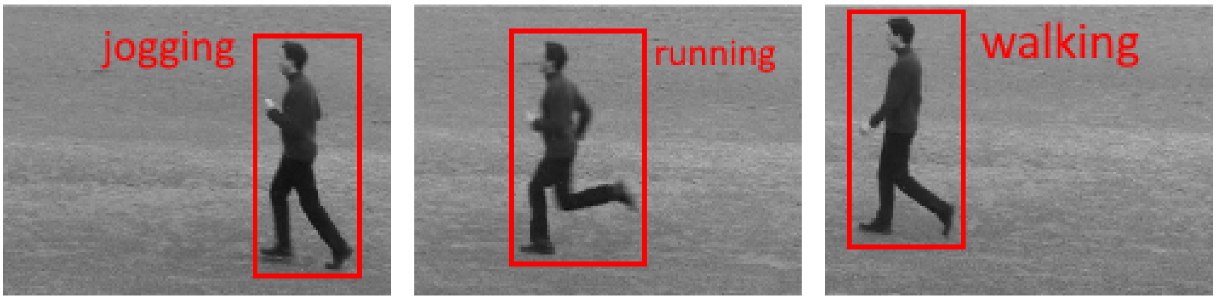
The following images illustrate structural keyframes sampled from the continuous sequences across the 6 action categories, which serve as the foundation for spatial-temporal analysis.

Figure 3: Input Dataset: Keyframes sampled from the 40-frame continuous sequences to enable dynamic kinematic processing.

7 Sample Target Output

The following graphics represent the expected visual output of the processing system, demonstrating successful subject localization via a dynamic bounding box and correct global semantic category labeling.

(a) Boxing (b) Handclapping (c) Handwaving



(d) Jogging (e) Running (f) Walking

Figure 4: Final Output Framework: Bounding box localization and global action classification label.